

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Computer Science & Engineering

2. Subject Name: Operating Systems

L-T-P: 3-1-0

Credits: 4

3. Pre-requisites: Computer Organization and Architecture

4. Syllabus and reference books:

Syllabus:

Introduction to OS; Concept of process, states and state transition; Scheduling algorithms, context switching; Process synchronization and inter-process communication; Threads, popular thread libraries, thread synchronization; Deadlock: necessary conditions, avoidance and prevention; Memory management: Contiguous and non-contiguous allocation, physical and logical addresses, paging, virtual memory; File systems: file operations, organization, mounting, sharing; File system implementation; Disk and I/O management; Virtualization: motivation and techniques; Case studies of operating systems.

Reference Books:

- 1) Operating Systems Concepts, by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne, Wiley India, 2018 or later
- 2) Modern Operating Systems, 4th edition, by Andrew Tanenbaum and Herveart Bos, Pearson, 2015 or later
- 3) Operating Systems: Three Easy Pieces, by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Arpaci-Dusseau Books LLC, 2018

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to OS	1
2.	Concept of process, states and state transition	3
3.	Scheduling algorithms, context switching	3
4.	Process synchronization and inter-process communication	5
5.	Threads, popular thread libraries, thread synchronization	3
6.	Deadlock: necessary conditions, avoidance and prevention	2
7.	Memory management: Contiguous and non-	3

	contiguous allocation, Physical and logical addresses	
8.	Memory management: Paging, different Page Table architectures, Virtual Memory	6
9.	File systems: file operations, organization, mounting, sharing	3
10.	File system implementation	3
11.	Disk and I/O management	3
12.	Virtualization: motivation and techniques	3
13.	Case studies of operating systems	2
Total number of hours		40